Fiscal Year:	FY 2004	**Task Last Updated:**	FY 06/01/2004
PI Name:	Matthiesen, David Ph.D.		
Project Title:	Diffusion Processes in Molten Semiconductors (DPIMS)		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Metals and alloys		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	dhm5@po.cwru.edu	**Fax:**	FY 216 368-8618
PI Organization Type:	UNIVERSITY	**Phone:**	216 368-1366
Organization Name:	Case Western Reserve University		
PI Address 1:	Materials Science and Engineering Department, 420 White Buil		
PI Address 2:	10900 Euclid Avenue		
PI Web Page:	http://dmseg5.cwru.edu/		
City:	Cleveland	**State:**	OH
Zip Code:	44106	**Congressional District:**	11
Comments:			
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No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	4
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA MSFC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
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Key Personnel Changes/Previous PI:			
COI Name (Institution):	Stroud, D. (Ohio State University) Chait, A. (NASA Glenn Research Center)		
Grant/Contract No.:	NAG8-152		
Performance Goal No.:	1.25		
Performance Goal Text:	Use microgravity to establish and improve quantitative and predictive relationships between the structure, processing, and properties of materials.		

Task Description:

This program of study is directed at the fundamental and applied issues pertaining to diffusion of species in the liquid state as driven by convection gradients (Fickian diffusion) and thermal gradients (Soret diffusion). The fundamental materials systems of interest for the near term study are the dilute binary systems of gallium (Ga), antimony (Sb), and a dilute ternary system of gallium and silicon in germanium (Ge). Systems of commercial interest for future study include the binary systems of germanium and silicon. This research program consists of three major components: an experimental measurement portion, a continuum numerical simulation portion and an atomistic numerical simulation portion.
The experimental measurement portion is designed to provide definitive measurements of the purely diffusive component of mass transfer in molten semiconductor systems. A shear cell technique will be used to directly measure the diffusion coefficients in semiconductor melts. For the Fickian diffusion case, isothermal measurements will be used to determine the diffusion coefficients on temperature, dopant type and column diameter. For the Soret diffusion case, measurements will be made in a thermal gradient. The objectives of the Diffusion Processes in Molten Semiconductors program are to provide purely diffusive experimental measurements of the isothermal and thermomigration diffusion coefficients of Ga, Sb and Si dopants in molten germanium with sufficient accuracy and precision to:

a) Differentiate between model predictions of the temperature dependence

b) Determine the effect of dopant size and type

c) Determine if a "wall effect" is present

d) Provide input to continuum and atomistic model development

e) Provide a preliminary evaluation of dopant-dopant interactions

Another objective is to develop a continuum numerical model of the germanium diffusion column, shear cell, cartridge and furnace for both earth-based and space-based experiments which accurately predicts the measured concentration profile as a function of distance in the diffusion column.

A final objective is to develop atomistic models which accurately predict:

a) The purely diffusive isothermal diffusion coefficient of a dopant in a molten semiconductor

b) The temperature dependence of the dopants in molten semiconductors, and which

c) Attempts to explain the "wall effect"

d) Develops new empirical potentials useful for predicting other diffusion and transport properties for other molten semiconductor systems. The fundamental mechanisms of mass diffusion in the liquid state are still unclear to the degree necessary for the prediction of diffusion of one species into another or even within itself. This observation is especially true with respect to the dependency of diffusion mechanisms on temperature and on concentration levels, as well as, on the dopant type. Present estimates of the diffusivity in molten semiconductors can typically provide an order of magnitude estimate only, without any information on their dependency on concentration levels and types, and on temperature and temperature gradients.

The availability of these data is of paramount importance for practical reasons as well. The relevancy of numerical modeling for the analysis and design of ground-based and space-based experiments is directly dependent upon the accuracy of the fundamental material properties used in simulations. These data are also important for the correct characterization and interpretation of experimental results from ground-based and space-based experiments.

The subject of how a mass of one species diffuses through a matrix of another species is, at the same time, both a very old and a very new research area. That this area can encompass the small, i.e. movement of electrons in a plasma, to the very large, i.e., the depletion of the global ozone layer, merely serves to emphasize the fundamental aspects of this subject. Most manufacturing technologies at some stage, rely on diffusion processes in the solid, liquid or gas.

The need for precise measurements of the diffusion coefficients in molten semiconductors has been repeatedly pointed out. These data are required both to interpret the experimental results from previous space-based (and earth-based) experiments and also to optimize newly envisioned experiments. Difficulties in experimental techniques and theoretical interpretations are cited for the lack of these data. This is a comprehensive program which addresses both of these issues.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Task Progress:

A comparative study on the effectiveness of two different methods to reduce buoyancy driven convection to obtain an intrinsic diffusion coefficient was investigated. Both methods used the shear cell technique and antimony doped molten germanium. In the first method, the effectiveness of microgravity was investigated. In the second method, a ground-based isothermal heat pipe furnace system was used to reduce the thermal gradients. The effect of capillary size was also investigated using four capillary diameters (3.0 mm, 2.0 mm, 1.5 mm and 1.0 mm). The antimony concentration in the solidified capillaries was characterized using Inductively Coupled Plasma – Mass Spectrometry (ICP-MS) and a measured diffusion coefficient (Dmeasured) was determined using the long-time solution to Fick’s 2nd Law. The data obtained was extrapolated to a ‘zero diameter’ capillary to yield an intrinsic diffusion coefficient (Dintrinsic).
The microgravity experiments returned a Dmeasured value for the two 3.0 mm capillaries of (5.0878 +/- 0.4687) x 10-4 cm2/s and (5.6472 +/- 0.5948) x 10-4 cm2/s, with a value of (3.6911 +/- 0.2563) x 10-4 cm2/s for the 1.5 mm capillary. Because of the capillary diameter dependence on Dmeasured, it was concluded that the microgravity results were contaminated with convection.

The Dintrinsic value for Sb-doped molten Ge at 1000 oC was found to be (1.2230 +/- 0.0624) x 10-4 cm2/s. It was found that there was no influence of capillary size. This implies that there was no contribution from buoyancy driven convection for the ground-based isothermal furnace results. This Dintrinsic value was extrapolated to 938 oC using 1) the Stokes-Einstein equation and 2) the Arrhenius equation. For the Stokes-Einstein equation: D(938) = 0.9670 x 10-4 cm2/s. For the Arrhenius equation: D(938) = 1.0334 x 10-4cm2/s with a range of (1.0242 to 1.0470) x 10-4cm2/s.

The extrapolated diffusion coefficient values was compared to the recalculated mean value of 0.522 x 10-4 cm2/s with a

	range of 0.240 to 0.693 x 10-4 cm2/s reported by Burton et al. Thus, the isothermal furnace measurements were slightly above this, however, the extrapolated Dintrinsic value at 1000 oC was more precise.
Bibliography Type:	Description: (Last Updated: 05/28/2004)
Dissertations and Theses	Keefer, L.A. "Isothermal Diffusion Coefficient Measurements of Antimony Doped Molten Germanium Using the Modified Shear Cell Technique" Ph.D. thesis Dept of Materials Science and Engineering Case Western Reserve University Jan-2004